Exp. No 04 **Date: 24- 07- 2026**

Fire and Gas Monitoring System Using MQ-2 and Flame Sensors

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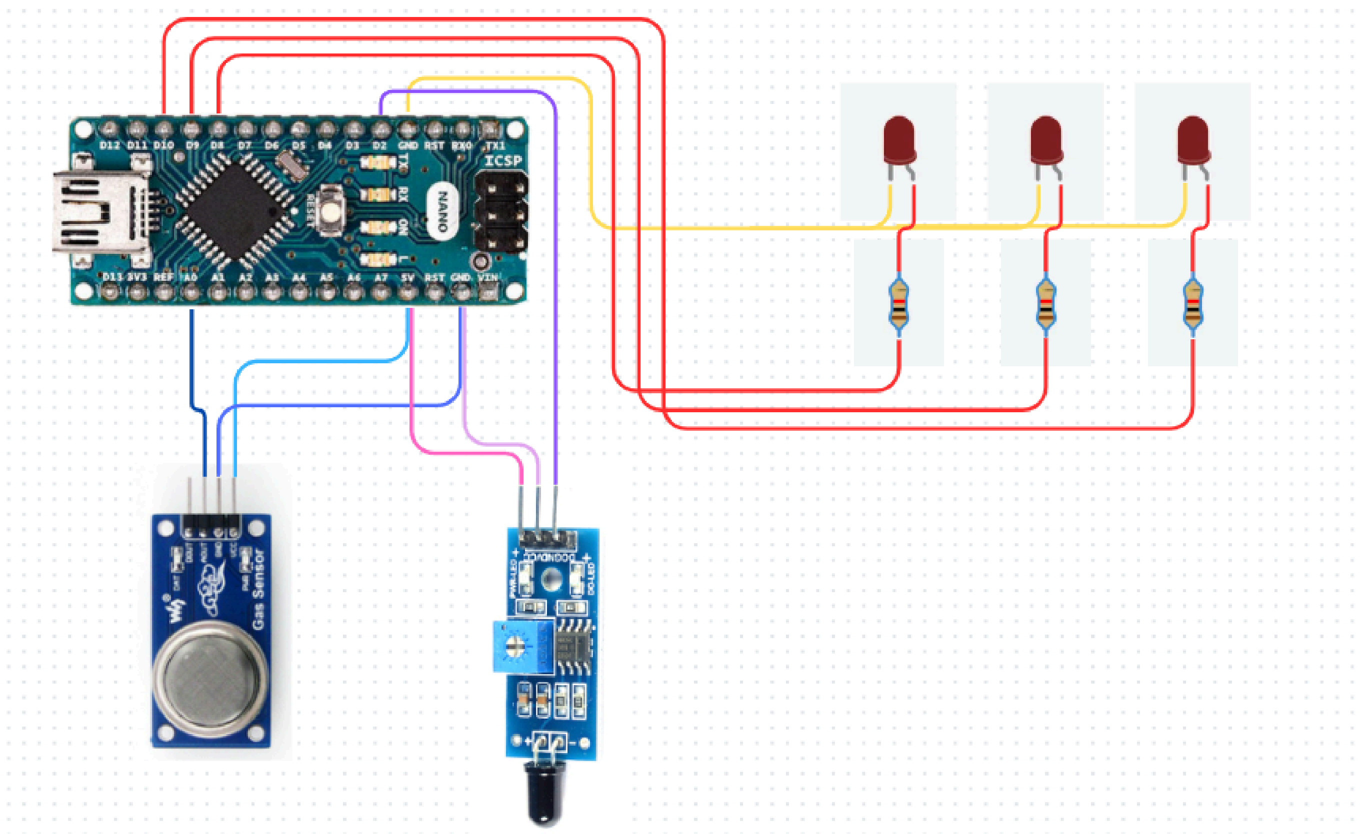
- ## Components Required

Sr.No.	Component	Quantity	Description	Use
1.	Arduino Nano	1	A small microcontroller board used for programming electronic projects.	Used to pass the code of detecting if gas, fire or none.
2.	Breadboard	1	A board with small holes used to make circuits.	Used to make the circuit easily and test connections.
3.	LED	3	Acts as the light.	Lights if fire ,gas or nothing is there.
4.	Resistor	3	A resistor is used to reduce the flow of current.	Used to protect the LED from excessive current.
5.	MQ-2 Sensor	1	A sensor that detects gas.	Used to detect gas.
6.	Flame Sensor	1	A sensor that detects fire.	used to detect if fire is there.

- ## Circuit Diagram

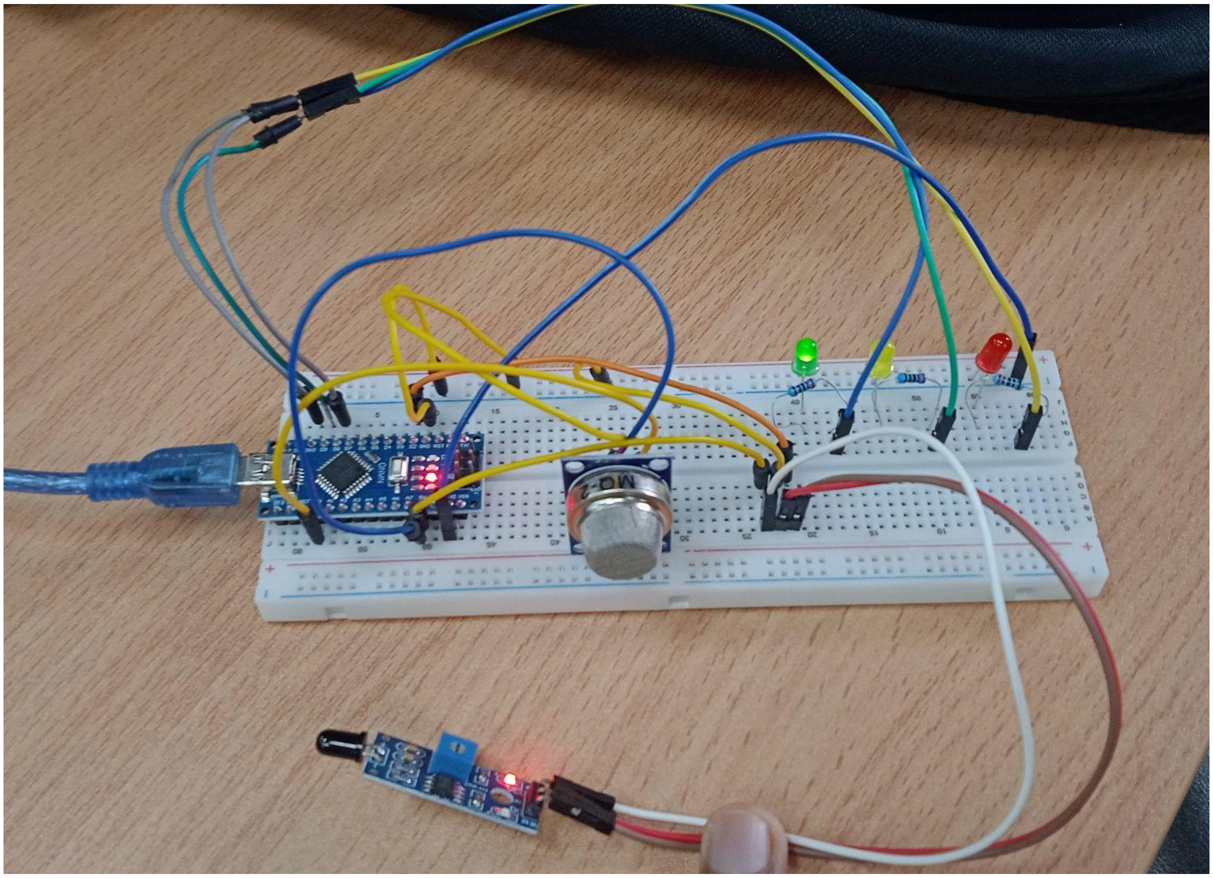


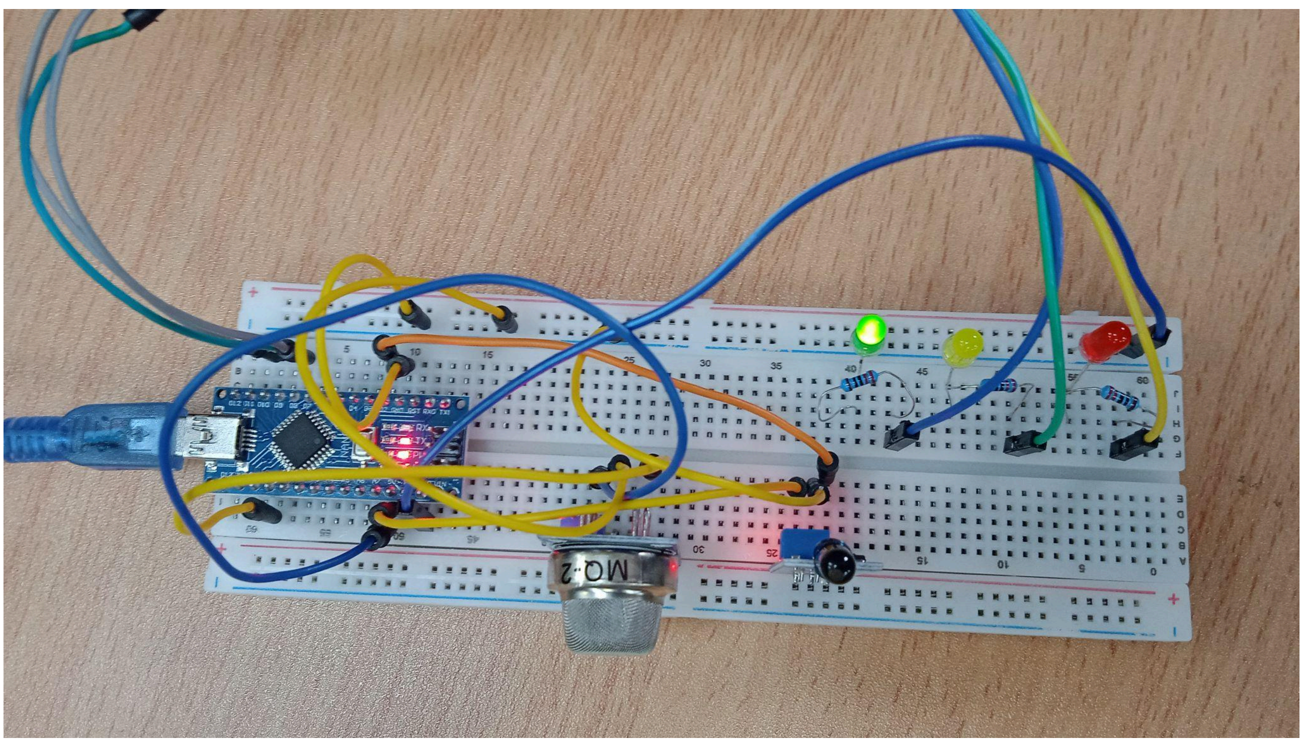
- **Program Code**

```
const int flamePin = 2;     // Flame sensor DO
const int gasPin = A0;      // MQ2 AO

const int fireLED = 8;
const int gasLED = 9;
const int normalLED = 10;

int gasThreshold = 300;     // Adjust after testing

void setup() {
  pinMode(flamePin, INPUT);

  pinMode(fireLED, OUTPUT);
  pinMode(gasLED, OUTPUT);
  pinMode(normalLED, OUTPUT);

  Serial.begin(9600);
}
```

```
void loop() {

  int flame = digitalRead(flamePin);
  int gasValue = analogRead(gasPin);

  Serial.print("Gas Value: ");
  Serial.print(gasValue);
  Serial.print("  Flame: ");
  Serial.println(flame);

  // Flame sensor outputs LOW when fire is detected
  if (flame == LOW) {
    digitalWrite(fireLED, HIGH);
    digitalWrite(gasLED, LOW);
    digitalWrite(normalLED, LOW);
  }
  else if (gasValue > gasThreshold) {
    digitalWrite(fireLED, LOW);
    digitalWrite(gasLED, HIGH);
    digitalWrite(normalLED, LOW);
  }
  else {
    digitalWrite(fireLED, LOW);
    digitalWrite(gasLED, LOW);
    digitalWrite(normalLED, HIGH);
  }

  delay(200);
}
```

- **Photograph of the Experiment**





- **Conclusion**

In this experiment we learned how to use the MQ-2 Gas Sensor and Flame Sensor with Arduino. We learned how these sensors detect gas and fire and to use LEDs to show different conditions like red LED glows when fire is detected, the yellow LED glows when gas is detected, and the green LED glows when everything is normal.